

# K-MEANS CLUSTERING

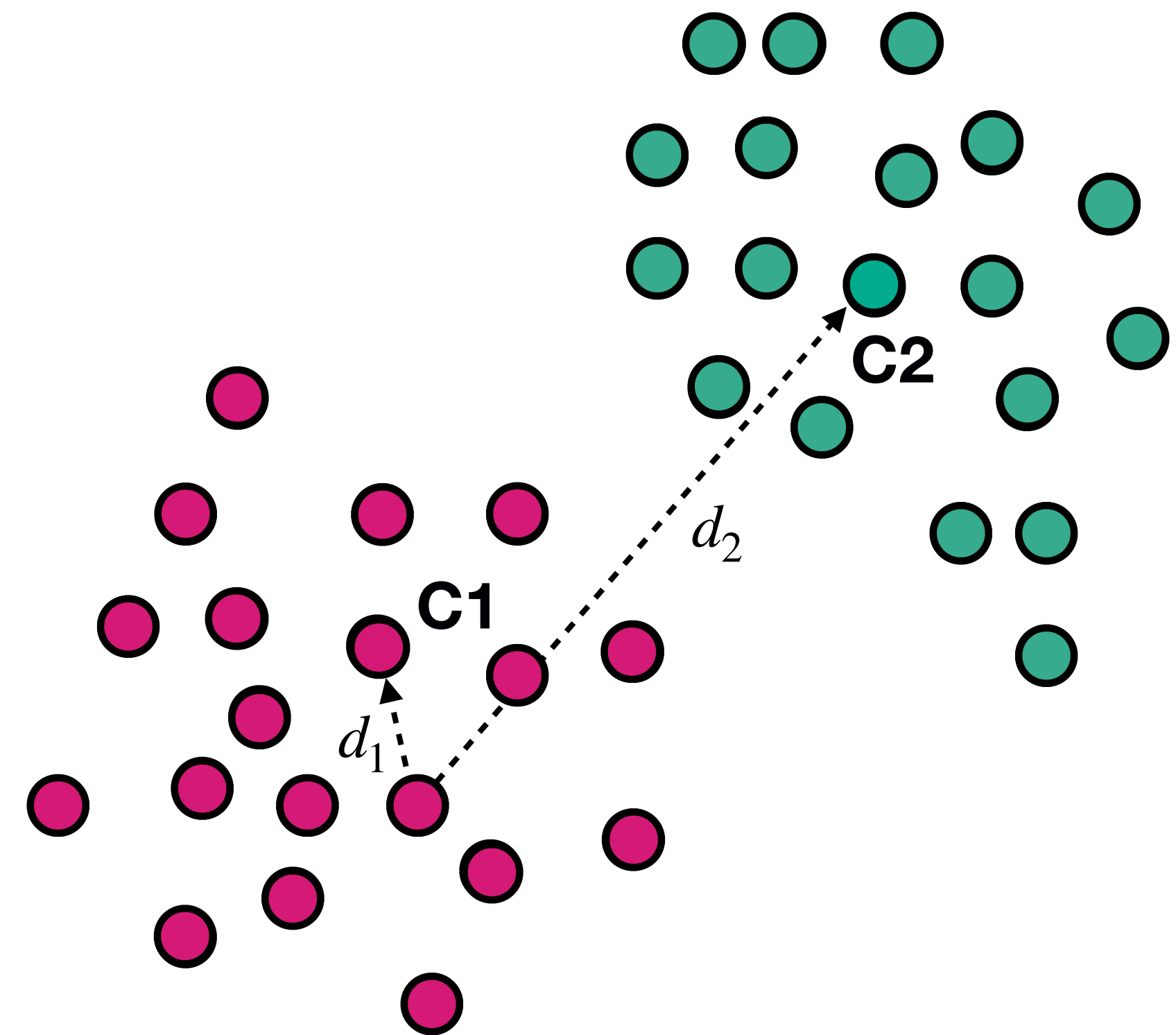
FROM NEIGHBOURHOODS TO GROUPING!

**Objective**– Partition  $n$  data points into  $K$  clusters such that the total within-cluster sum of square (WCSS) is minimized. Formally, the objective is to find:

$$\arg \min_S \sum_{i=1}^k \sum_{x \in S_i} ||x - \mu_i||^2^*$$

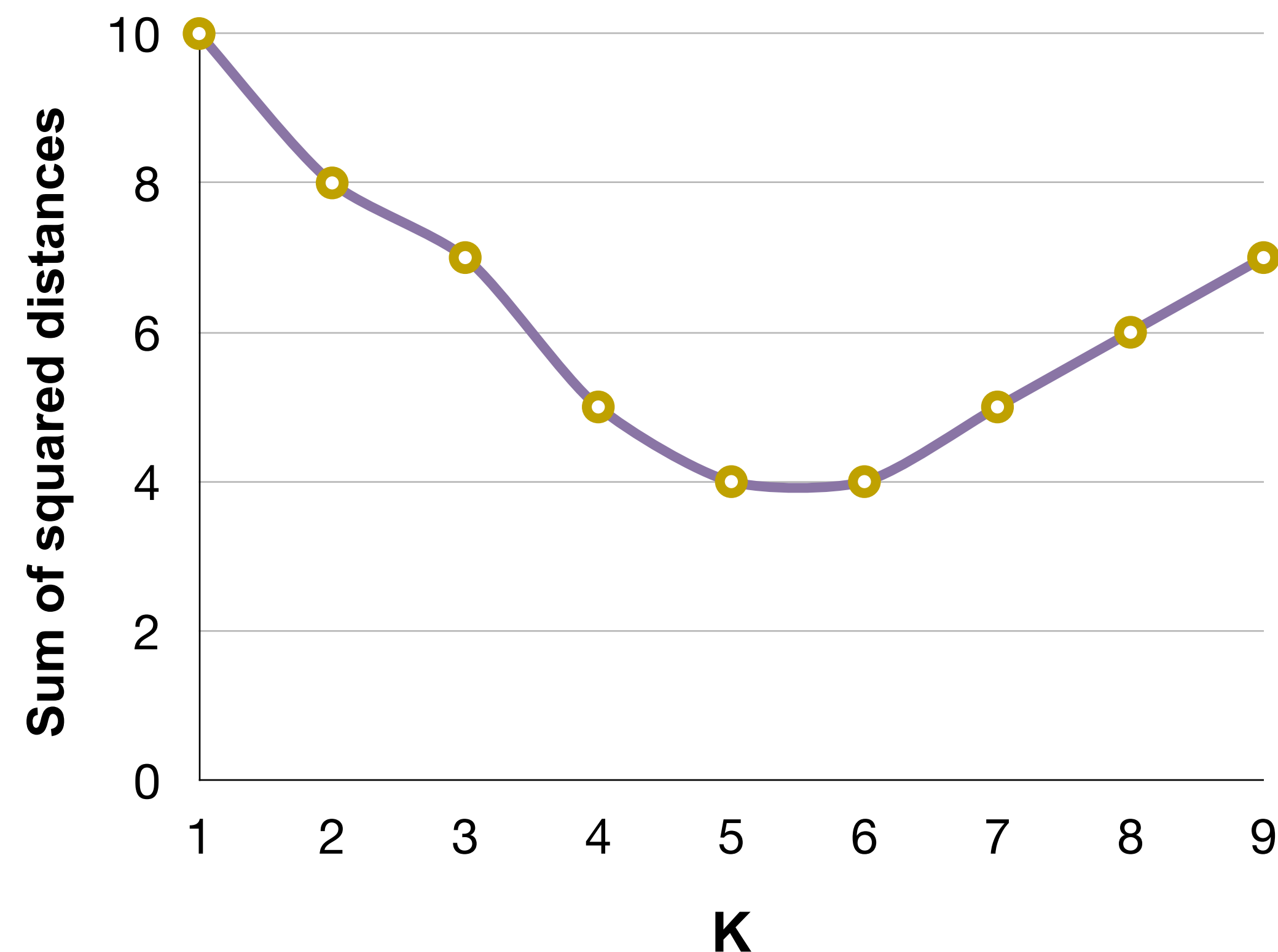
Where:

- $S$  are the  $K$  clusters.
- $\mu_i$  is the centroid or mean of all the points in  $S_i$ .



\* The  $|| \cdot ||$  notation in formulas often denotes a norm, which is a function that assigns a positive length or size to each vector in a vector space, excluding the zero vector which is always assigned a length of zero.

# SELECTING THE OPTIMAL $K$ IN KNN



## Elbow Method

- ▶ Compute the sum of squared distances from each point to its assigned center.
- ▶ Plot these values for a range of  $K$  values.
- ▶ Look for an "elbow" in the plot, a point where the rate of decrease sharply changes. This can indicate a good estimate for  $K$ .

The idea is that, after a certain point, adding more clusters will not provide much better fit to the data.